

Project 1: What is your project's name? \*

AI for server security

Project 1: What is your project description? \*

Sample project description - “As a technical intern with us, you'll collaborate on developing our existing startup ideas into Minimum Viable Products (MVPs). These MVPs will then be put to test! For this, you'll be working with a technical coach - usually a masters or PhD in computer science or data science. A significant focus of our projects lies in Generative AI; for instance, we've previously created AI models adept at interpreting food and drink orders.”

As a SuperSafe intern, you will collaborate on pen testing our Apache reverse proxy server & Python-based API server (to at least simulate D-DOS attacks), parse server log files, and develop AI-driven automations to enhance server security. Experience with a programming language is nearly necessary. Proficiency in Python would be greatly helpful. Familiarity operating in Linux environments would be nice to have. We are open to circumstances whereby interns develop these skills on-the-job. Our background in education has helped students complete projects from zero previous coding experience, although students must be motivated self-learners, willing to dive into the tutorials and resources we suggest.

Project 1: What would be the final output (MVP code, PPT, 5-pager report, website etc).? \*

At a minimum, the final output should be a presentation. More ideally, the student will be a co-author on a whitepaper, scholarly article, or conference proceeding (depending on the results).

**Project 1: What is the difficulty level of the project and what skill sets would the interns need to start work on this project? \***

We will use this information to match suitable interns to you so please be as specific as you'd like to be, especially if this is a technical project. For technical projects, please do specify the coding languages (example - Python, HTML, JAVA, PHP etc) + proficiency level (example - no coding experience needed/ beginner/advanced) that the interns will need to already know! Please also mention which is a hard-requisite and which are desired skill sets.

Experience with a programming language is nearly necessary. Proficiency in Python would be greatly helpful. Familiarity operating in Linux environments would be nice to have. We are open to circumstances whereby interns develop these skills on-the-job. Our background in education has helped students complete projects from zero previous coding experience, although students must be motivated self-learners, willing to dive into the tutorials and resources we suggest.

**Project 2: What is the project name? \***

Sample project name (for a project we run with another company) - "Building an MVP of a start-up Idea"

Enhanced UI for Quantum-Safe Windows App

**Project 2: What is the project description? \***

Sample project description - “Building an MVP of a start-up Idea: As a technical intern with us, you'll collaborate on developing our existing startup ideas into Minimum Viable Products (MVPs). These MVPs will then be put to test! For this, you'll be working with a technical coach - usually a masters or PhD in computer science or data science. A significant focus of our projects lies in Generative AI; for instance, we've previously created AI models adept at interpreting food and drink orders.”

As a SuperSafe intern, you will collaborate to improve the user-interface (UI) of our quantum-safe secure file transfer Windows application. The program is written in Python. Therefore, having either Python experience or Visual Studio (.NET) experience would be very helpful. A simpler version of this project would be to enhance the UI still purely in Python. A possibly more involved project could consist of developing a skeleton application in Visual Studio (i.e., an aesthetic but non-functional prototype).

**Project 2: What would be the final output (MVP code, PPT, 5-pager report, website etc).? \***

MVP code.

**Project 2: What is the difficulty level of the project and what skill sets would the interns need to start work on this project? \***

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Having either Python experience or Visual Studio (.NET) experience would be very helpful.

### Project 3: What is the project name? \*

Sample project name (for a project we run with another company) - "Building an MVP of a start-up Idea"

Learn Secure Coding AI Chat Bot

### Project 3: Project Description \*

Sample project description - "Building an MVP of a start-up Idea: As a technical intern with us, you'll collaborate on developing our existing startup ideas into Minimum Viable Products (MVPs). These MVPs will then be put to test! For this, you'll be working with a technical coach - usually a masters or PhD in computer science or data science. A significant focus of our projects lies in Generative AI; for instance, we've previously created AI models adept at interpreting food and drink orders."

As a SuperSafe intern, you will collaborate to develop an AI large language model (LLM) chat bot web application for learning best practices for coding secure applications, especially proper API key management.

Project 3: What would be the final output (MVP code, PPT, 5-pager report, website etc).? \*

MVP code.

### Project 3: What is the difficulty level of the project and what skill sets would the interns need to start work on this project? \*

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HTML/CSS/JS experience would be very helpful with NextJS experience preferred. However, a vanilla JS app could be fine, as well as using ReactJS, RedwoodJS, Vue, Angular, etc. JS frameworks. PHP-based apps, Java, Python-based e.g., using Django, etc. etc. are also acceptable. We are open to circumstances whereby interns develop these skills on-the-job. Our background in education has helped students complete projects from zero previous coding experience, although students must be motivated self-learners, willing to dive into the tutorials and resources we suggest.

### Project 4: Name \*

Sample project name (for a project we run with another company) - "Building an MVP of a start-up Idea"

### Project 4: Project Description \*

Sample project description - "As a technical intern with us, you'll collaborate on developing our existing startup ideas into Minimum Viable Products (MVPs). These MVPs will then be put to test! For this, you'll be working with a technical coach - usually a masters or PhD in computer science or data science. A significant focus of our projects lies in Generative AI;

for instance, we've previously created AI models adept at interpreting food and drink orders.”

As a SuperSafe intern, you will collaborate in the development of a quantum-safe electronic health record system. As such, a major focus will be utilizing cryptographic methods to properly store and retrieve sensitive data.

**Project 4: What would be the final output (MVP code, PPT, 5-pager report, website etc).? \***

MVP Code.

**Project 4: What is the difficulty level of the project and what skill sets would the interns need to start work on this project? \***

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Experience with a programming language is nearly necessary. Familiarity with Python is preferred. We are open to circumstances whereby interns develop these skills on-the-job. Our background in education has helped students complete projects from zero previous coding experience, although students must be motivated self-learners, willing to dive into the tutorials and resources we suggest.

Project 5: Project Name \*

Project 5: Project Description \*

Sample project description - "Building an MVP of a start-up Idea: As a technical intern with us, you'll collaborate on developing our existing startup ideas into Minimum Viable Products (MVPs). These MVPs will then be put to test! For this, you'll be working with a technical coach - usually a masters or PhD in computer science or data science. A significant focus of our projects lies in Generative AI; for instance, we've previously created AI models adept at interpreting food and drink orders."

As a SuperSafe intern, you will be analyzing mental health data sets with AI/ML methods to enhance mental health characterization and aid with the decision-making process of clinical depression treatment options. The goal of this project is to develop an MVP analysis pipeline in close collaboration with our partner who is finishing his MD at Stanford and about to begin an MBA at Harvard.

Project 5: What would be the final output (MVP code, PPT, 5-pager report, website etc).? \*

MVP code.

Project 5: What is the difficulty level of the project and what skill sets would the interns need to start work on this project? \*

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Coding experience is nearly necessary. Experience with Python is preferred. Familiarity with any (or of course all) of the following would be quite helpful: data analysis in general, AI/ML techniques in general, and PyTorch in particular.